Cross Modal Point Cloud Completion with Geometric Transformation

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Abstract: Point cloud completion faces the persistent challenge of reconstructing high-fidelity, detail-rich point clouds with minimal computational cost. Unlike traditional unimodal methods, view-guided point cloud completion (VIPC) offers a promising alternative way by leveraging low-cost 2D images as supplementary information to enhance reconstruction details. This article introduces a deep learning network for the VIPC task by employing geometric priors, which we call VGTNet, meaning view-guided geometric transformation network. Our method consists of a CGTNet, which learns geometric transformations and predicts a coarse point cloud skeleton of missing regions, and a Refine Module that subsequently upsamples and refines this point skeleton. Compared with the traditional strategy of pure encoder-decoder based methods, our method can preserve more geometric details for generation. By effectively integrating the intrinsic geometric information of the point cloud with the supplementary structural cues provided by external 2D images, our method achieves high-quality global shape reconstruction and preserves fine-grained local details.

Keywords: Point Cloud Completion, Multi-modal Fusion, Geometric Priors

1. Introduction

Point clouds, a commonly used 3D data representation, play an essential role in numerous applications [1], [2]. Devices like LiDAR and depth cameras capture precise point clouds representing portions of object and scene surfaces. However, even after merging multiple partial scans, the resulting point clouds often remain unevenly distributed and incomplete [3], which limits the advancement of 3D-related applications.

To address this issue, Point Cloud Completion (PCC) has become increasingly important. Conventional PCC methods rely on model-based predictions for missing parts, but their performance remains limited [3]. Driven by advances in artificial intelligence and deep learning, most state-of-the-art methods now employ fully supervised learning [4], [5], [6], [7], [8]. Given the challenge of obtaining ground-truth (GT) point clouds in practice, there is also growing interest in self-supervised [9], [10], [11] and weakly supervised approaches [12], [13]. Moreover, inspired by the success of generative models, particularly diffusion models [14], recent works explore these as promising next-generation solutions for shape completion [15], [16], [17], [18].

On the contrary, many studies have explored fusing point clouds with images [19]. The View-Guided PCC (VIPC) task [20] is a cross-modal point cloud completion problem. Early methods [20], [21] followed traditional PCC pipelines [7], using 2D information to refine point clouds. Recent works adopt more sophisticated fusion strategies that can achieve strong results without explicit point refinement. For example, XMFNet [12] streamlined the task through an efficient fusion framework, while EGIINet [22] leveraged Vision Transformers (ViT) [23] and Meta-Transformer [24] to unify point clouds and images into a shared token representation. The latest approach, MAENet [25], combines a two-stage coarse-to-fine process with feature pyramid fusion, achieving state-of-the-art performance and improved point distribution uniformity. However, this comes at the cost of a large parameter count and reduced computational efficiency.

PCC can be reformulated as a spatial geometric transformation task. This approach avoids the information loss common in encoder–decoder architectures and enables more accurate detail reconstruction [26]. Recent studies have leveraged such geometric priors for shape completion. GTNet [27] employs spatial operations such as symmetry, rotation, translation, and scaling to recover missing regions. More recently, SymmCompletion [26] incorporated partial symmetry priors to enhance the robustness of geometric transformation methods for asymmetric objects. Motivated by these insights, we hypothesize that the rich structural and semantic information in 2D images can also facilitate the learning of more accurate geometric transformations for point cloud completion.

Therefore, in this paper, we propose a View-guided Geometric Transformation Network (VGTNet) that leverages image-guided geometric priors. Rather than directly predicting the entire shape, our method uses local geometric priors to infer the geometry of missing regions. Since the transformation is flexible and not restricted to simple mirroring, it can also handle asymmetric objects effectively. A subsequent refinement module further enhances geometric details and ensures uniform point distribution by utiliz-

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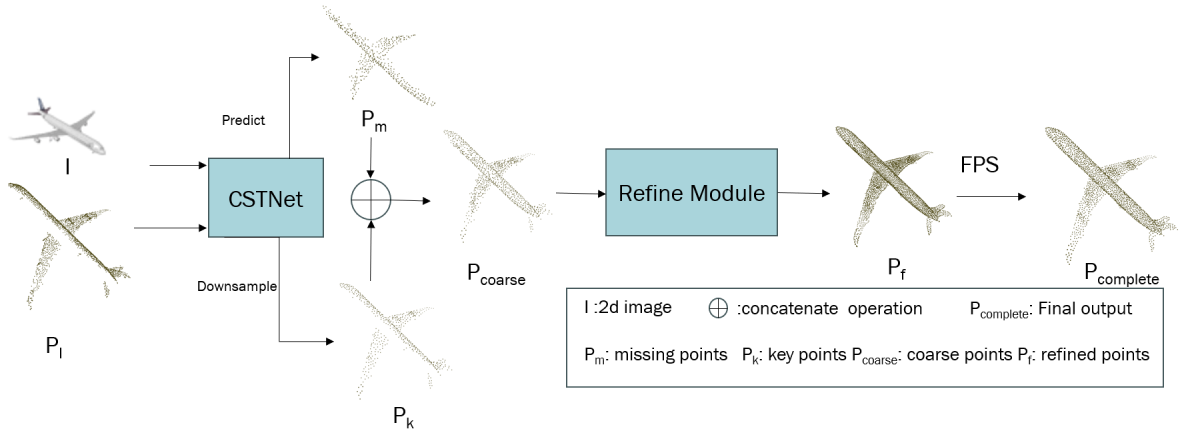


Fig. 1 An illustration of our proposed two-stage framework. By giving a uncomplete sparse point cloud and its correspond 2D image in the dataset, the model first employs the CGTNet for generating the initial coarse point cloud by predicting the missing geometry. Subsequently, followed Refinement Module enhances this coarse output, producing a detailed and uniform point cloud with a standard density (2048 points).

ing rich texture and structural features extracted from the input image.

Our method's unique contributions are summarized as follows:

(1) To the best of our knowledge, we are the first to employ geometric transformations to enhance the cross-modal point cloud completion task.

(2) Our VGTNet comprises two specialized modules: a transformation prediction network (CGTNet), which learns to generate missing point clouds by applying learned transformations to the partial input, and a refinement network, which globally optimizes the completed shape for detail and uniformity.

Through experiments, we demonstrate that our algorithm achieves state-of-the-art performance on the ShapeNet-VIPC dataset benchmark [20] while maintaining computational efficiency. The results also show that our method outperforms all competitors in terms of generalization ability.

2. Related work

2.1 3D Point Cloud Completion

3D point cloud completion (PCC) aims to recover high-fidelity, complete, and evenly distributed point clouds from sparse and incomplete inputs. Early PCC approaches predominantly followed a coarse-to-fine pipeline. Pioneering this field, PCN [7] introduced a two-stage framework: first generating a coarse but structurally complete point cloud skeleton, then refining it with a decoder [28] to restore fine-grained details in the final output. This foundational design has since been adopted and improved by numerous subsequent methods [4], [29].

More recently, an alternative strategy has emerged that preserves the original structure of the partial input. Methods such as the Transformer-based Pointr [6] focus solely on predicting the missing regions, which are then merged with the known points. While this approach effectively maintains the input geometry, it suffers from a key limitation: mismatched point density between the original and predicted regions often results in visible holes, gaps, and non-uniformity, thereby degrading the quality of the reconstructed shape [26].

2.2 VIPC task

Recently, multimodal fusion has been attracting increasing attention [30], [31], [32]. Images are easy to obtain and can serve as valuable supplementary information for point cloud analysis. In this context, the cross-modal, image-guided PCC task was introduced by VIPC [20], which also provided the first benchmark dataset for this problem — ShapeNet-VIPC [20]. The initial VIPC model follows the coarse-to-fine pipeline proposed by PCN [7], first using a Transformer for single-view reconstruction and then fusing this with the partial input before an image-supported final refinement stage.

Subsequent studies have explored various cross-modal fusion strategies for this task. XMFNet [12] employs multiple layers of self-attention and cross-attention mechanisms to effectively fuse image and point cloud features in the latent space, achieving strong results without a dedicated refinement module. EGIINet [22] adopts an explicit information interaction strategy, tokenizing both modalities to create a highly parameter-efficient model. More recently, MAENet [25] applies a multi-layer feature pyramid for fusion, followed by a refinement network to ensure uniform point distribution. However, the refinement stage also incurs a high computational cost.

2.3 Use of Geometric Transformation

Leveraging geometric priors is a powerful approach in 3D shape completion, with symmetry being a particularly effective cue [27]. Such priors have been successfully applied in various 3D vision tasks [9], [33]. In PCC research, GTNet [27] reconceptualized the completion problem as a geometric transformation task. Rather than directly generating points, it learns transformations—such as rotation, translation, and scaling—to replicate existing object parts and fill in missing regions. Because these transformations are flexible and not restricted to simple mirroring, the approach is also effective for asymmetric objects. Building on this idea, the recent SymmCompletion method [26] uses local symmetry priors extracted from local features. By focusing on local symmetry, it extends applicability to complex, asym-

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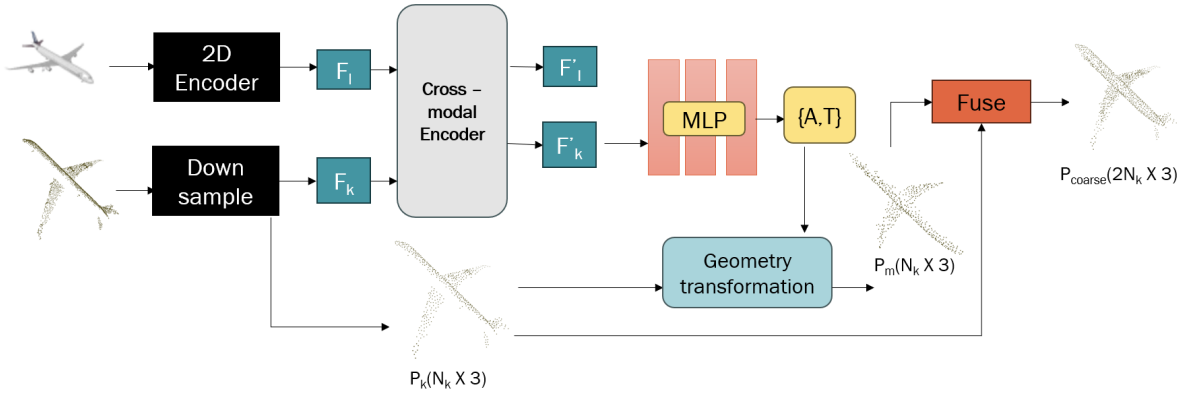


Fig. 2 The main architecture of our Cross Geometric transformation Network (CGTNet). This network first extracts parallel features from the image and point cloud using intra-modal encoders. These features are then fused via a cross-modal attention mechanism. Finally, a prediction head (MLP) uses the fused features to regress the parameters of a geometric transformation, which is applied to input keypoints to generate the missing geometry.

metric shapes and serves as a key inspiration for our work. Our method aims to incorporate these geometric transformation concepts into a view-guided completion framework.

3. Proposed Method

3.1 Overview

Our VGTNet addresses the view-guided completion task through a two-stage architecture. The core objective is to reconstruct a high-fidelity point cloud $\boldsymbol{P}_{\text{complete}} \in \mathbb{R}^{N_q \times 3}$ from the input sparse partial point cloud $\boldsymbol{P}_{\text{in}} \in \mathbb{R}^{N_p \times 3}$ and a single-view image $\boldsymbol{I} \in \mathbb{R}^{H \times W \times C}$, which serves as additional information for guidance.

Our framework decouples this challenge into two distinct subproblems: coarse structure completion and image-guided refinement module. The overall pipeline is shown in Fig. 1.

1. Cross Geometric transformation Network (CGTNet) focuses on generating a complete but coarse representation of the object's geometry. Instead of directly predicting the entire shape, CGTNet leverages local geometic priors to infer the geometry of the missing regions. The predicted missing geometry $\boldsymbol{P}_{\text{m}}$ is then merged with a downsampled set of keypoints $\boldsymbol{P}_{\text{k}}$ to form a coarse point cloud, denoted as $\boldsymbol{P}_{\text{coarse}}$.

2. Refinement Module enhances geometric details and ensures the uniform distribution of $\boldsymbol{P}_{\text{coarse}}$. Importantly, this module leverages rich texture and structural features extracted from the additional input image $\boldsymbol{I}$ to guide the refinement process, correcting inaccuracies and adding fine details that might be missed by the preceding CGTNet stage. Effectively combining the inherent structural information of the 3D point cloud with the supplementary cues from the 2D image remains the primary challenge at this stage.

Finally, to produce the output with a standardized point count, the refined point cloud follows a downsampling operation using Farthest Point Sampling (FPS) [34], a very common downsampling method. This step ensures the final output $\boldsymbol{P}_{\text{complete}}$ is a complete, detailed, and uniformly distributed point cloud with N_c points. The specific implementations of CGTNet and the Refinement Module are detailed in the subsequent sections.

3.2 Cross Geometric Transformation Network (CGTNet)

The Cross Geometric transformation Network (CGTNet), which is illustrated in Fig. 2, serves as the foundational module in our framework. Its primary function is to coarse point cloud which is nearly complete by learning an image-guided geometric transformation that exploits geometric priors. The architecture of CGTNet can be broken down into three main stages: intra-modal encoding, cross-modal fusion, and transformation-based point generation. We will now introduce these three modules in detail.

3.2.1 Dual-Stream Intra-Modal Encoding

We begin by extracting features from each modality in parallel.

- **Image Feature Extraction:** For the image stream, we employ a lightweight yet powerful ResNet18 [35] as the image encoder to extract a feature map $\boldsymbol{F}_I$ containing rich semantic and textural information.
- **Point Cloud Feature Extraction:** For the geometry stream, we process the input point cloud $\boldsymbol{P}$ using a hierarchical downsampling architecture detailed in Fig. 3. This process, which utilizes Set Abstraction layers from PointNet++ [36] followed by a Point Transformer [37], yields two critical outputs: a set of downsampled keypoints $\boldsymbol{P}_{\text{k}}$ that form a sparse skeleton of the object, and their corresponding high-dimensional features $\boldsymbol{F}_{\text{k}}$.

3.2.2 Cross-Modal Feature Fusion

To enrich the geometric features with visual context from the image, we fuse the features from both streams. Following recent successful approaches [12], [22], [38], we use a cross-attention mechanism to enable 2D and 3D information fusion between $\boldsymbol{F}_{\text{I}}$ and $\boldsymbol{F}_{\text{k}}$. To enhance this interaction, the attention layers share parameters, a technique noted for its effectiveness in [22]. The fusion process is defined as:

$$\boldsymbol{F}'_{\text{I}} = \gamma(\boldsymbol{F}_{\text{I}}, \boldsymbol{F}_{\text{k}}), \tag{1}$$

$$\boldsymbol{F}'_{\text{k}} = \gamma(\boldsymbol{F}_{\text{k}}, \boldsymbol{F}_{\text{I}}), \tag{2}$$

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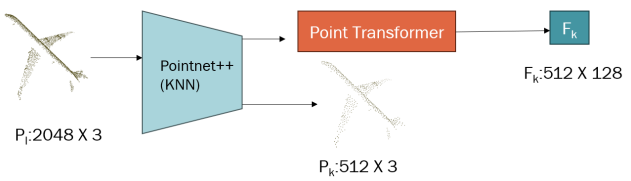


Fig. 3 Detailed architecture of the point cloud down-sampling module. This module employs PointNet++ style Set Abstraction layers followed by a Point Transformer to extract hierarchical features while producing a sparse set of keypoints.

where γ represents the combined cross-attention and self-attention block, and $\mathbf{F}'_{\mathrm{I}}$ and $\mathbf{F}'_{k}$ are the resulting cross-modal features.

3.2.3 Coarse Point Generation

Inspired by GTNet [27] and SymmCompletion [26], we formulate the prediction of the missing points as a spatial transformation problem. Instead of directly regressing point coordinates, we use the fused keypoints features $\boldsymbol{F}'_{\mathrm{k}}$ to predict the parameters of an affine transformation. Two prediction heads, implemented as lightweight MLPs, are used for this purpose:

$$\boldsymbol{P}_{\mathrm{m}} = \boldsymbol{P}_{\mathrm{k}} \cdot \{\mathcal{X}(\boldsymbol{F}'_{\mathrm{k}})\} + \mathcal{Y}(\boldsymbol{F}'_{\mathrm{k}}). \tag{3}$$

Here, $\mathcal{X}(\cdot)$ and $\mathcal{Y}(\cdot)$ are the MLP heads that predict the transformation matrix $\mathbf{A} \in \mathbb{R}^{N_k \times 3 \times 3}$ and the translation vector $\mathbf{T} \in \mathbb{R}^{N_k \times 3}$, respectively.

Finally, the coarse point cloud $\boldsymbol{P}_{\mathrm{coarse}}$ is formed by concatenating the original keypoints and the newly generated missing points: $\boldsymbol{P}_{\mathrm{coarse}} = \boldsymbol{P}_{\mathrm{k}} \cup \boldsymbol{P}_{\mathrm{m}}$. This complete but coarse shape is then passed to the refinement module.

3.3 Image-Guided Refinement Module

The coarse point cloud $\boldsymbol{P}_{\mathrm{coarse}}$ generated by CGTNet provides a complete geometric structure but may lack fine details and uniform point distribution. The purpose of the Refinement Module, inspired by recent advances in the field [26], [39], [40], is to address these issues by leveraging a rich set of multi-source information. As depicted in Fig. 4, this module takes four distinct feature sets as input: the coarse point features ($\boldsymbol{F}_{\mathrm{coarse}}$), the keypoint features ($\boldsymbol{F}_{\mathrm{k}}$), the predicted missing point features ($\boldsymbol{F}_{\mathrm{m}}$), and the global image features ($\boldsymbol{F}_{\mathrm{I}}$). The details of the lightwise decoder to process the information from pre stage can be seen in Fig.6. How to effectively fuse features from so many different sources is one of the main challenges of this module.

3.3.1 Multi-Source Information Fusion

The core of this module is a hierarchical fusion pipeline designed to inject both visual and geometric guidance into the coarse features.

- **Image Cross-modal Guidence:** To leverage the 2D image, we first fuse the image 2D modal features $\boldsymbol{F}_{\mathrm{I}}$ with both the coarse initial point cloud features $\boldsymbol{F}_{\mathrm{coarse}}$ and the predicted missing part points features $\boldsymbol{F}_{\mathrm{m}}$. This can be achieved by using a cross-attention mechanism where 2D image serves as the context (key/value) for the point features (query), thereby enriching the geometric representations with detailed visual cues from the image.
- **Geometric Self-Guidance:** Next, to ensure the refined shape respects the original known geometry and the predicted structure, referring to [26]'s method, we use the "parts" to guide the "whole". Specifically, we do fusion operation to the fused coarse features $\boldsymbol{F}'_{\mathrm{coarse}}$ with both the fused keypoint features $\boldsymbol{F}'_{\mathrm{k}}$ and the fused missing point features $\boldsymbol{F}'_{\mathrm{m}}$. This allows the high-confidence information from the original partial scan ($\boldsymbol{F}_{\mathrm{k}}$) and the structural information from the predicted part ($\boldsymbol{F}_{\mathrm{m}}$) to guide the refinement of the overall shape.

This multi-level fusion process, employing combined cross-attention and self-attention blocks ($\mathcal{T}, \mathcal{Q}$), culminates in a highly informed feature representation, $\boldsymbol{F}''_{\mathrm{coarse}}$, for the coarse point cloud:

$$\boldsymbol{F}''_{\mathrm{coarse}} = \left[\mathcal{T}(\boldsymbol{F}'_{\mathrm{coarse}}, \boldsymbol{F}'_{\mathrm{k}})\,;\, \mathcal{Q}(\boldsymbol{F}'_{\mathrm{coarse}}, \boldsymbol{F}'_{\mathrm{m}})\right], \tag{4}$$

where $\mathbf{F}'$ denotes features that have been enriched by the image context, and $[\cdot\,;\,\cdot]$ represents feature concatenation. The detailed architecture of our fusion layer design can be checked in Fig. 5.

3.3.2 Decoder and Detail Enhancement

The final stage translates the enhanced features into a high-fidelity point cloud. First, the fused feature $\boldsymbol{F}''_{\mathrm{coarse}}$ is passed through two consecutive self-attention layers, denoted by $\zeta(\cdot)$, for final feature propagation and enhancement.

Instead of a standard deconvolutional decoder, we adopt an efficient upsampling strategy based on learned point displacements. We first duplicate the points in $\boldsymbol{P}_{\mathrm{coarse}}$ and then predict a 3D offset for each point using a lightweight MLP that takes the enhanced features as input. This process is formulated as:

$$\boldsymbol{P}\mathrm{f} = \boldsymbol{P}\mathrm{up} + \Delta\boldsymbol{P} \tag{5}$$

where

$$\boldsymbol{P}\mathrm{up} = \mathcal{U}(\boldsymbol{P}\mathrm{coarse}) \quad \text{and} \quad \Delta\boldsymbol{P} = \mathbf{MLP}(\zeta(\boldsymbol{F}''\mathrm{coarse})). \tag{6}$$

$\boldsymbol{P}_{\mathrm{up}}$ denote the upsampled point cloud. $\Delta\boldsymbol{P}$ is the offset of every points.

To conclude, and in line with methods like MAENet [25], this dense cloud is downsampled to the standard 2048 points. This upsampling-then-downsampling strategy has been shown to be effective at producing a final point cloud with a highly uniform distribution.

3.4 Loss Function

Loss function is very important in our training process, the training of our network is guided by a composite loss function, $\mathcal{L}_{\mathrm{total}}$, which is designed to enforce both geometric fidelity in the reconstructed point cloud and meaningful fusion between the 2D and 3D modalities. Specifically it consists of two components: Point Completion loss and cross-modal style transfer loss.

3.4.1 Point Completion Loss

To ensure the geometric accuracy of the output, we employ the Chamfer distance implemented by L1 norm regularization (CD_{l1}), a classic and widely used metric. It can be defined as follows:

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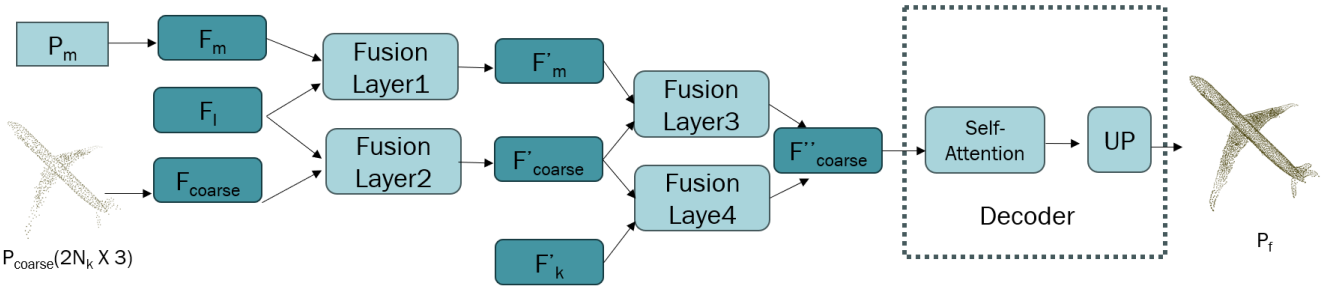


Fig. 4 Overview of the Refinement Module architecture. The module first processes the coarse point cloud and predicted missing points using parallel Lightwise Encoders to extract distinct feature sets. These features, along with image and keypoint features, are then progressively fused through a series of identical Fusion Layers to produce the final refined shape.

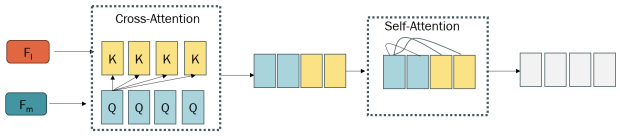


Fig. 5 Detailed structure of a single Fusion Layer. This layer combines a cross-attention block for inter-modal information exchange (e.g., image features as key/value, point features as query) with a self-attention block for intra-modal feature enhancement.

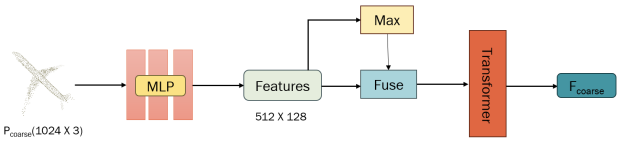


Fig. 6 The detailed architecture of the Lightwise Encoder. It processes a point set by first using MLPs for point-wise feature extraction, followed by a max-pooling operation and a Transformer block for contextual feature aggregation.

$$d_{CD}(\boldsymbol{P_1}, \boldsymbol{P_2}) = \frac{1}{|\boldsymbol{P_1}|} \sum_{n_1 \in P_1} \min_{n_2 \in P_2} \|n_1 - n_2\|_2, + \frac{1}{|\boldsymbol{P_2}|} \sum_{n_2 \in P_2} \min_{n_1 \in P_1} \|n_2 - n_1\|_2. \quad (7)$$

Here, $\boldsymbol{P_1}$, $\boldsymbol{P_2}$ stands for the two sets of points to be compared. In order to give full utilize the role of each model stage, a key aspect of our training strategy is application of multi-stage supervision. Instead of only supervising the final output, we apply the CD loss at multiple stages of our pipeline to ensure accuracy is maintained throughout the generation process. Our total completion loss, $\mathcal{L}_{\text{CD}}$, is the sum of these individual losses:

$$\mathcal{L}_{\text{CD}} = d_{\text{CD}}(\boldsymbol{P}_{\text{coarse}}, \boldsymbol{Q}) + d_{\text{CD}}(\boldsymbol{P}_{\text{f}}, \boldsymbol{Q}) + d_{\text{CD}}(\boldsymbol{P}_{\text{complete}}, \boldsymbol{Q}) \quad (8)$$

$\boldsymbol{Q}$ is the ground truth point cloud, as the terms $\boldsymbol{P}_{\text{coarse}}$, $\boldsymbol{P}_{\text{f}}$, $\boldsymbol{P}_{\text{complete}}$ respectively represent the outputs from the CGTNet, the Refinement Module, and the final downsampling step, respectively. This approach provides direct gradients to each major component of our network.

3.4.2 Style Transfer Loss

To encourage effective feature fusion between the 2d image and 3d point cloud modalities, we incorporate a style transfer loss, which has proven effective in recent cross-modal methods [22], [38]. This loss enforces consistency in the second-order statistics (i.e., style) of the feature distributions before and after the cross-modal interaction and thus enhance the quality of information fusion. It is formulated as:

$$\mathcal{L}_{\text{style}} = \frac{\|G(F_i) - G(F'_p)\|_F^2 + \|G(F_p) - G(F'_i)\|_F^2}{n_p \times C}. \quad (9)$$

Here, $\boldsymbol{F}_{\text{p}}$ and $\boldsymbol{F}_{\text{i}}$ are the intra-modal features in the CGTNet process, while $\boldsymbol{F}'_{\text{p}}$ and $\boldsymbol{F}'_{\text{i}}$ are the features after cross-modal fusion. $G(\cdot)$ is the Gram matrix function, defined as $G(\boldsymbol{F}) = \boldsymbol{F}^T \boldsymbol{F}$.

3.4.3 Overall Loss

Our overall loss function is constructed by aggregating the point completion loss and the style transfer losses, with appropriate weighting to balance their contributions:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CD}} + \alpha \cdot \mathcal{L}_{\text{style}}, \quad (10)$$

where α is a hyperparameter that balances the contribution of each term.

4. Experiments

4.1 Dataset

The entirety of our experimental evaluation is carried out on the **ShapeNet-VIPC** dataset [20].It contains 8 types of objects for normal test and 4 unseen types of objects for generation ability test. All input settings and evaluation metrics remain exactly the same as previous methods. Each object in the dataset is associated with 24 partial point clouds, each generated from a unique viewpoint and paired with a corresponding 2D image, simulating real-world occlusion scenarios. To realize a fair and direct comparison with prior method, we adhere strictly to the data splits method and experimental protocols used in [12], [20], [21], [22], [25], [38].

4.2 Evaluation Metrics

We evaluate the ability of our current model by using two standard and wide used metrics: L2 norm regularized Chamfer distance (l_2-CD) [41] together with F-Score [42]. The two matrics are formulated as follows:

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$$d_{CD}(P_1, P_2) = \frac{1}{|P_1|} \sum_{n_1 \in P_1} \min_{n_2 \in P_2} \|n_1 - n_2\|_2^2, + \frac{1}{|P_2|} \sum_{n_2 \in P_2} \min_{n_1 \in P_1} \|n_2 - n_1\|_2^2, \tag{11}$$

$$F\text{-Score}(i) = \frac{2 \cdot \text{Pre}(i) \cdot \text{Re}(i)}{\text{Pre}(i) + \text{Re}(i)}. \tag{12}$$

Here Pre means precision, Re means recall. In general, the L2 CD distance measures the detail quality of the completed point cloud, while the F-Score measures the overlap between the completed point cloud and the Ground Truth point cloud.

4.3 Implementation Details

The input point cloud has been initially preprocessed to a standard size, 2048 points via Farthest Point Sampling (FPS) [34],image is 224×224 low resolution RGB. Final network output is our predict complete point cloud, it also contains 2048 points, matching the ground truth density.

Within our architecture, the CGTNet module first downsamples the input point cloud to 512 keypoints. It then generates an additional 512 points, forming a 1024-point coarse cloud. The Refinement Module subsequently upsamples this coarse cloud by a factor of 4 to a dense 4096-point cloud, which is finally downsampled via FPS to the standard 2048-point-resolution output.

All models were developed and trained by using PyTorch [43] and CUDA [44]. Batch size is set 64 for 200 training epochs. We found that the best optimal hyperparameter settings were different for different categories, but for most categories (including the training of the model in the generalization experiment) , the learning rate is set as 0.0005, α in total loss is set as 0.03.

4.4 Results and Analysis

In our experiment, We conducted a comprehensive test and evaluation of our proposed model on eight representative categories from the Shapenet-VIPC dataset [20]. Our method was benchmarked against a suite of latest approaches.

The quantitative results, which are presented in Tab 1 and Tab 2, show the performance superiority of our network. Our method achieves the best CD-L2 and F-score performance on most categories. In some categories, such as "lamp", the performance is much better than other competitors, and a few indicators are slightly inferior to MAENet [25], but the gap is very small. This demonstrates the superior performance of our method over other state-of-the-art methods. We visualized some objects in Fig7 and Fig8 for easy comparison. The comparison methods all used the pre-trained models provided for inference. To show the details of generated points, we zoomed in on the details. The point cloud generated by our method is richer in details than that of other methods and is closer to the GT point cloud while we keep the whole uniform density of the point cloud.

To test the generalization ability of the model, we conducted experiment on 4 unknown categories in the dataset and compared it with the state-of-the-art methods, results can be found in Tab. 3. Our method surpasses all competitors in generalization ability and achieves state-of-the-art performance.

We also visualized our results and made visual comparisons. Since the author of MAENet did not provide a pre-trained model, we only compared it with EGIINet. The visual comparison results can be found in Fig. 9. It can be seen that our method has better quality.

5. Conclusion and Limitations

In this paper, we proposed a network——VGTNet, which is designed to address the cross-modal point cloud completion task (VIPC task) that effectively leverages geometric priors. Our network is a two-stage architecture which firstly generates a coarse structure by learning image-guided geometric transformations, and then progressively refines it to produce a high-fidelity, uniformly distributed output. The experiments validate our approach and validate that our model performs best on the Shapenet-VIPC benchmark [20] compared with MAENet [25] and EGIINet [22].

For the limitation of our method, although our VGTNet demonstrates certain progress in the performance from the test results, the use of the refinement module will inevitably increase the computational cost. We compared our approach with competing methods in terms of model size; the results are available in the Tab4. Compared to EGIINet's [22] lightweight token design, our approach struggles to achieve a comparable model size due to the inclusion of a refine module. However, compared to the currently top-performing MAENet [25], our approach achieves significant advantages in model size and performs better, demonstrating its data efficiency.

Possible optimization directions include lightweighting of the model. Another further research plan in the future is to use generative models or large language models(LLM) to assist in the generation of point clouds.

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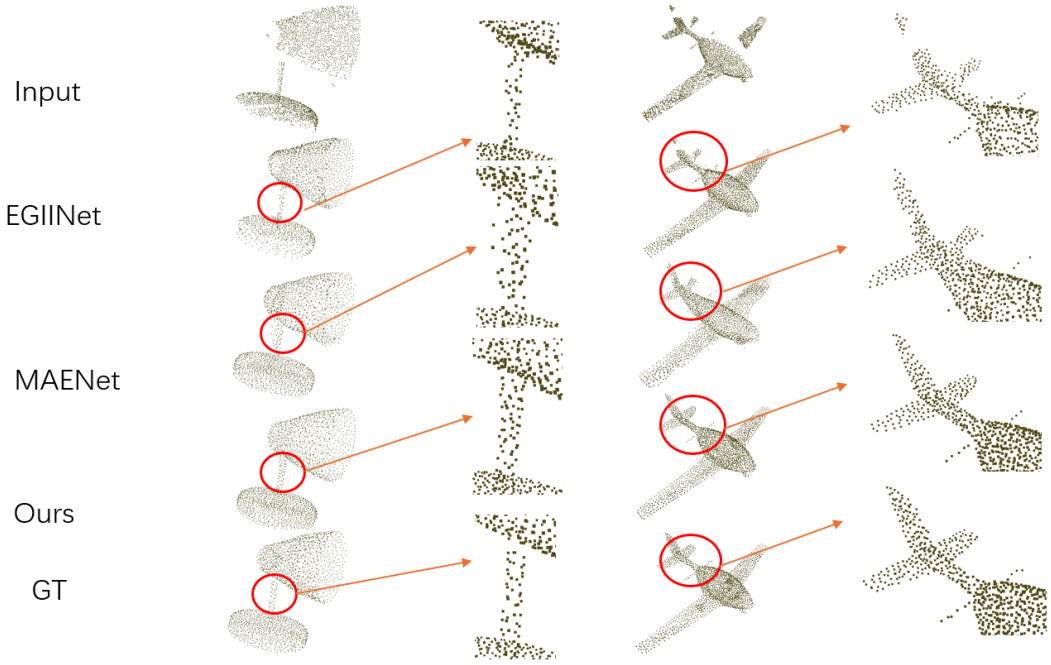


Fig. 7 In a visual comparison of the lamp and plane categories, our method achieves a smoother transition on the neck of the lamp than our competitors. For the airplane object, our method is closer to the GT in recovering details of the fuselage and tail of the rear half of the aircraft.

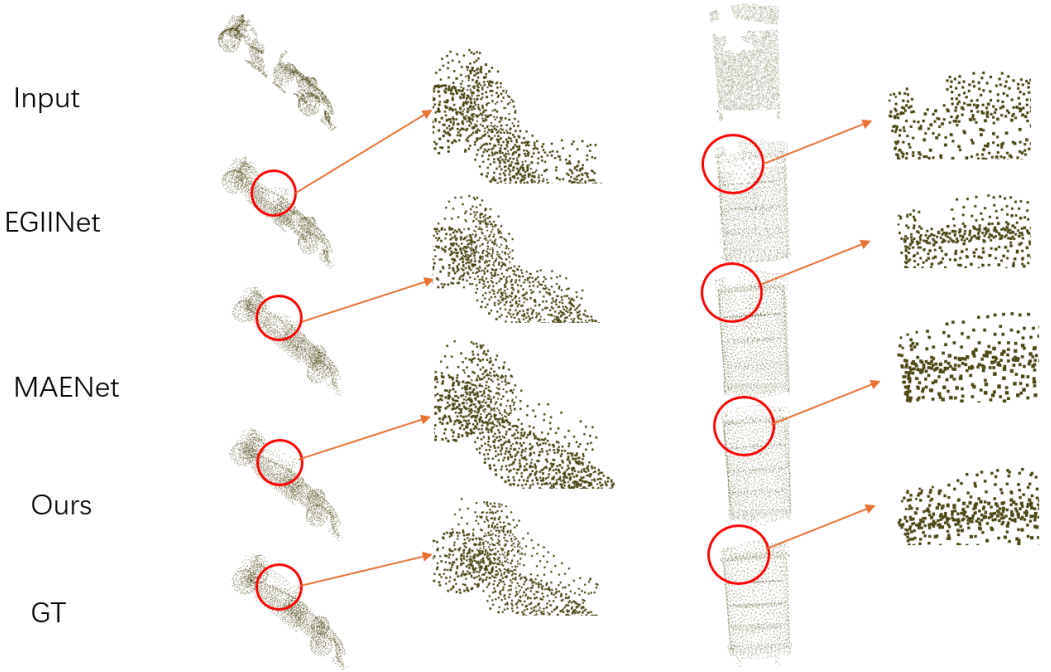


Fig. 8 In a visual comparison of the car and cabinet categories,compared with MAENet and EGIINet, our method better restores the upper part of the sports car and better completes the missing part in the top of the cabinet, which is closer to the result of GT.

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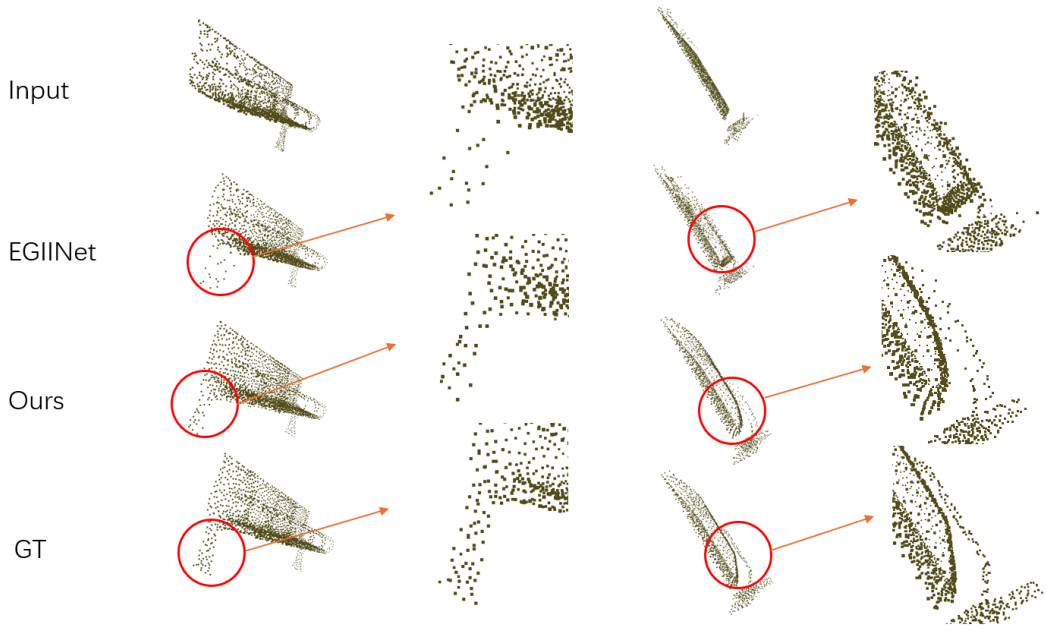


Fig. 9 Visual comparison of model generalization capabilities. On the bench, our method reproduces the details of the bench leg better than EGIINet. On the monitor object, our method reproduces the monitor stand well, while EGIINet fails to reproduce it successfully.

Table 1 Comparision of Mean Chamfer Distance per point (CD $\times 10^3 \downarrow$).

Methods	Airplane	Cabinet	Car	Chair	Lamp	Sofa	Table	Watercraft	Avg
PCN [7]	4.246	6.409	4.840	7.441	6.331	5.668	6.508	3.510	5.619
PoinTr [6]	1.686	4.001	3.203	3.111	2.928	3.507	2.845	1.737	2.851
PointAttN [5]	1.613	3.969	3.257	3.157	3.058	3.406	2.787	1.872	2.853
ViPC [20]	1.760	4.558	3.183	2.476	2.867	4.481	4.990	2.197	3.308
CSDN [21]	1.251	3.670	2.977	2.835	2.554	3.240	2.575	1.742	2.570
XMFNet [12]	0.572	1.980	1.754	1.403	1.810	1.702	1.386	0.945	1.443
EGIINet [22]	0.534	1.921	1.655	1.204	0.776	1.552	1.227	0.802	1.211
MambaTron [38]	0.537	1.905	1.620	1.184	0.774	1.561	1.234	0.779	1.199
MAENet [25]	0.483	1.608	1.524	1.053	0.624	1.317	**1.034**	0.687	1.041
VGTNet(Ours)	**0.438**	**1.589**	**1.460**	**1.018**	**0.552**	**1.309**	1.037	**0.648**	**1.006**

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Table 2 Mean F-Score @ 0.001 ↑.

Methods	Airplane	Cabinet	Car	Chair	Lamp	Sofa	Table	Watercraft	Avg
PCN [7]	0.578	0.270	0.331	0.323	0.456	0.293	0.431	0.577	0.407
PoinTr [6]	0.842	0.516	0.545	0.662	0.742	0.547	0.723	0.780	0.683
PointAttN [5]	0.841	0.483	0.515	0.683	0.729	0.512	0.699	0.774	0.662
ViPC [20]	0.803	0.451	0.512	0.529	0.706	0.434	0.594	0.730	0.591
CSDN [21]	1.251	3.670	2.977	2.835	2.554	3.240	2.575	1.742	0.695
XMFNet [12]	0.961	0.662	0.691	0.809	0.792	0.723	0.830	0.901	0.796
EGIINet [22]	0.969	0.693	0.723	0.847	0.919	0.756	0.857	0.927	0.836
MambaTron [38]	0.969	0.697	0.728	0.851	0.920	0.764	0.856	0.931	0.839
MAENet [25]	0.981	0.754	0.756	0.885	0.945	**0.818**	**0.895**	0.951	0.873
VGTNet(Ours)	**0.988**	**0.756**	**0.774**	**0.888**	**0.953**	0.813	0.894	**0.957**	**0.878**

Table 3 Results on unknown categories of ShapeNet-Vipc Dataset (CD × 10^3 ↓, F-score @ 0.001↑).

Methods	Avg		Bench		Monitor		Speaker		Cellphone	
	CD ↓	F-score ↑	CD ↓	F-score ↑	CD ↓	F-score ↑	CD ↓	F-score ↑	CD ↓	F-score ↑
PoinTr [6]	3.755	0.619	1.976	0.797	4.084	0.599	5.913	0.454	3.049	0.627
PointAttN [5]	3.674	0.605	2.135	0.764	3.741	0.591	5.973	0.428	2.848	0.637
ViPC [20]	4.601	0.498	3.091	0.654	4.419	0.491	7.674	0.313	3.219	0.535
CSDN [21]	3.656	0.631	1.834	0.798	4.115	0.598	5.690	0.485	2.985	0.644
XMFnet [12]	2.671	0.710	1.278	0.862	2.806	0.677	4.823	0.556	1.779	0.748
EGIInet [22]	2.354	0.750	1.047	0.902	2.513	0.716	4.282	0.598	1.574	0.784
MAENet [25]	1.387	0.839	0.870	0.935	1.276	0.852	2.411	0.656	0.989	0.911
VGTNet(Ours)	**1.243**	**0.859**	**0.692**	**0.962**	**1.148**	**0.874**	**2.185**	**0.682**	**0.949**	**0.917**

Table 4 Comparisons on model sizes

Methods	Number of parameters (M)
ViPC [20]	17.43
CSDN [21]	16.85
XMFnet [12]	9.57
EGIINet [22]	**9.03**
MAENet [25]	32.72
Ours	20.80

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